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tikzpicture[ i=stealth, arr/.style = -i, thick, arrb/.style = i-i, thick, box/.style = rectangle, draw,
rounded corners=3pt, minimum width=1.8cm, minimum height=0.60cm, font=, align=center, fill=white,
sensor/.style = box, fill=orange!15, draw=orange!70, percep/.style = box, fill=blue!12, draw=blue!60, in-
jury/.style = box, fill=red!10, draw=red!55, fsmbox/.style = box, fill=purple!12, draw=purple!60, minimum
width=1.8cm, minimum height=3.8cm, ctrl/.style = box, fill=green!12, draw=green!60!black, expctrl/.style
= box, fill=cyan!12, draw=cyan!60, web/.style = box, fill=yellow!10, draw=yellow!60!black, hw/.style =
box, fill=gray!15, draw=gray!60, lbl/.style = font=, ]
[sensor] (orb) at ( 0.0, 1.0) Orbbec
; [sensor] (rs) at ( 0.0, -1.0) RealSense
;
[percep] (yolo) at ( 2.8, 2.2) YOLO-Pose; [percep] (kf) at ( 2.8, 1.0) Kalman 3D; [injury] (inj) at ( 2.8,
-0.1) Injury
; [percep] (surf) at ( 2.8, -1.4) Surface
;
[fsmbox] (fsm) at ( 5.8, 0.5) Mission
;
[ctrl] (wbc) at ( 8.8, 2.2) WBC QP; [expctrl] (exp) at (8.8, 1.0) Exp. Scanner; [ctrl] (mux) at ( 8.8, -0.2)
IK Goal
; [ctrl] (imp) at ( 8.8, -1.4) Impedance
;
[web] (webui) at ( 8.8, -2.4) Web UI
;
[hw] (spot) at (11.8, 1.0) Spot
; [hw] (z1) at (11.8, -0.6) Z1 arm
;
[arr] (orb.east) -- (yolo.west); [arr] (yolo.south) -- (kf.north); [arr] (rs.east) -- (surf.west); [arr] (rs.north)
-- node[above left,font=]RGB (inj.west);
[arr] (kf.east) -- node[above,font=] $\hat{p}$  (fsm.west -- kf); [arr] (inj.east) -- node[above left,font=]detect.
(fsm.west -- inj); [arr] (surf.east) -- node[above,font=] $p_s, n_s$  (fsm.west -- surf);
[arr] (fsm.east -- wbc) -- node[above,font=]en. (wbc.west); [arr] (fsm.east -- exp) -- node[above,font=]en.
(exp.west); [arr] (fsm.east -- mux) -- node[above,font=]sel.(mux.west); [arr] (fsm.east -- imp) -- node[above,font=]en.
(imp.west);
[arr] (wbc.south) -- (mux.north); [arr] (exp.south) -- (mux.north);
[arrb] (fsm.south east) -- node[below,font=,sloped]click/state (webui.north west);
[arr] (wbc.east) -- node[above,font=,yshift=2mm] $v_x, \omega_z$  (spot.west);
[arr] (mux.east) -- node[above,font=,yshift=2mm] $p^*$  (z1.west);
[arr] (imp.east) -- node[above,font=] $\tau$  (z1.west);
[lbl] at ( 0.0, 2.8) Sensors; [lbl] at ( 2.8, 3.0) Perception; [lbl] at ( 5.8, 2.9) Mission mgmt.; [lbl] at ( 8.8,
3.0) Controllers; [lbl] at (11.8, 2.2) Hardware;

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